

Generation Count and Residual Parameters in the Standard Model Representation Regime

A Residual-Exhaustion Theorem for Fixed-Boundary Data

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Abstract

This paper proves a residual-exhaustion theorem for generation count and residual parameters within the fixed Standard Model representation regime. Working downstream of prior representation-family, charge-lattice, and fixed-load closure results, the analysis asks which residual degrees of freedom can still be tightened by admissibility-preserving same-scope operations. The answer is a finite classification: residual candidates are either excluded, identified as redundant, mechanically forced by coherence, classified as presentation-level skin, or retained as fixed-boundary data.

The result does not claim an internal derivation of the numerical value three for fermion generations, nor does it claim a same-scope derivation of all gauge couplings, Yukawa values, mixing angles, phases, or scale-setting constants. Instead, it proves that any purported same-scope derivation of these residuals must factor through the terminal residual quotient already determined by the admissible classification. Attempts to force additional values internally collapse into hidden-selector import, witness enlargement, envelope change, skin reparameterization, or violation of standing preservation.

Generation count is therefore classified as a fixed-boundary cardinal datum of the Standard Model representation regime, while non-skin residual numerical parameters are classified as externally fixed residual data rather than arbitrary free knobs. The theorem is compatible with enlarged-domain physical theories, including ultraviolet completions or boundary-condition frameworks, provided they change the comparison regime rather than pretending to complete the original same-scope structure. The paper thus establishes classification rigidity, not metaphysical finality: within the declared representation regime, no further admissible same-scope residual tightening remains.

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1 Introduction and theorem target

The target question is not whether the observed Standard Model has three generations, whether the numerical values of its couplings and masses are empirically known, or whether larger-domain ultraviolet programs may attempt to explain those values. The target question is prior and more sharply typed:

Within the fixed Standard Model representation envelope already reached by SM-FINAL and SM-PT-1–SM-PT-3, can any remaining generation-count, multiplicity, or numerical-parameter freedom be further tightened by a same-scope admissible operation without changing the envelope, importing a selector, or violating standing preservation?

The answer proved here is no. This is not a declaration that no broader theory can derive additional numerical data. It is a residual classification theorem. Once the same-scope comparison class is fixed, every alleged remainder has one of five statuses:

EXCLUDE, IDENTIFY, FORCE, Skin, EXT-FIX.

No sixth same-scope status remains; the fifth status is written EXT-FIX and denotes externally fixed residual data.

The argument proceeds by constructing the residual comparison class explicitly. It supplies a fixed residual scope, a witness-complete comparison class, local definitions of the AASC tests, hidden-selector obstruction, residual reconstruction category, finite toy witness, route-exhaustion ledger, and interpretation-boundary discipline. The resulting theorem does not rest on an informal assertion that the remaining quantities are simply left over. It identifies the admissible terminal status of each residual class and proves that a proposed same-scope tightening operation must either factor through that status map, collapse to presentation, fail admissibility, or change scope.

Core claim. Let S_{res} be the fixed residual scope obtained downstream of SM-FINAL and SM-PT-1–SM-PT-3. Every admissibility-bearing residual candidate in S_{res} either fails maximality, identifies as redundant presentation, is forced by the declared global-coherence witness package, is skin, or is externally fixed. Generation cardinality and standing-bearing residual numerical values fall into the externally-fixed residual-data class unless an enlarged envelope supplies new anchors or tensor witnesses. No admissible same-scope operation remains that can bind them internally.

Main sequent. For the declared residual scope,

$$\begin{aligned} & \text{Adm}(S_{\text{res}}), \text{Stand}(X), W_{\text{res}}(X), T_{\text{res}}(X), \text{RPC}(X) \\ \implies & \tau(X) \in \{\text{EXCLUDE}, \text{IDENTIFY}, \text{FORCE}, \text{Skin}, \text{EXT-FIX}\}. \end{aligned}$$

If a proposed binder B attempts to force N_g or a residual parameter value while preserving all components of S_{res} , then

$$B \in \{\text{representation-equivalent}, \text{skin}, \text{selector import}, \text{repair}, \text{scope change}\},$$

and hence B is not a same-scope residual-tightening operator.

2 Scope lock and non-claims

The theorem is conditional and fixed-domain. It classifies residuals inside a declared Standard Model representation regime and a fixed comparison class. It does not range over arbitrary richer theories unless those theories preserve the same skeleton, witnesses, transport relations, quotient identities, and envelope ledger.

Definition 2.1 (Fixed residual scope). *A fixed residual scope is a tuple*

$$S_{\text{res}} = (K_{\text{SM}}, R_{\text{SM}}, T_{\text{SM}}, W_{\text{res}}, \sim_{\text{st}}, \mathcal{E}_{\text{SM}}),$$

where:

- (i) K_{SM} is the fixed Standard Model representation skeleton: field-role slots, representation-family roles, charge-lattice data, interaction slots, anomaly/witness ledger, and family-typing already licensed by upstream closure.
- (ii) R_{SM} is the set of admissible same-scope redescrptions.
- (iii) T_{SM} is the licensed transport family preserving standing-bearing identity across admissible descriptions.
- (iv) W_{res} is the residual witness package: multiplicity witnesses, family cardinality records, parameter slot records, quotient identities, boundary matching records, and coherence constraints still relevant after SM-PT-3.
- (v) $\sim_{\text{st}}$ is the standing equivalence relation induced by admissible redescription and witness preservation.
- (vi) $\mathcal{E}_{\text{SM}}$ is the envelope ledger: declared boundary data, generation cardinality, residual numerical parameter slots, normalization conventions, empirical calibration locations, and any larger-domain imports not derived internally by the same-scope theorem.

Definition 2.2 (Same-scope residual competitor). *A proposal P is a same-scope residual competitor only if it preserves S_{res} up to admissible equivalence. In particular, it must preserve $K_{\text{SM}}, R_{\text{SM}}, T_{\text{SM}}, W_{\text{res}}, \sim_{\text{st}}$, the no-selector/no-repair discipline, and the envelope boundary $\mathcal{E}_{\text{SM}}$. A proposal that adds a sterile sector, new gauge factor, altered Higgs sector, changed anomaly ledger, new flavor witness, new boundary selector, new metric, new typicality measure, or new value-selection rule changes scope unless it factors back through the same residual quotient.*

Non-claims. This manuscript does not claim:

- a derivation of the observed number three for fermion generations inside the fixed same-scope residual theorem;

- a derivation of numerical gauge-coupling values, Yukawa eigenvalues, Higgs-sector parameter values, CKM/PMNS numerical coordinates, CP residues, or scale-setting values;
- a prohibition on BSM, UV, flavor, or boundary-condition programs that enlarge the envelope and then attempt further value fixation;
- that empirical calibration is illegitimate physics;
- that externally-fixed values are arbitrary or meaningless;
- uniqueness across unconstrained mathematical imagination.

It claims only fixed-scope residual exhaustion: after the declared Standard Model representation envelope is fixed, no additional admissible tightening operation remains within that same scope.

Remark 2.3 (Interpretation boundary). *The results below establish rigidity and exhaustion properties of admissible same-scope residual classification within the fixed Standard Model representation regime. They do not, by themselves, assert metaphysical inevitability of the observed numerical constants, nor do they deny the possibility of physically meaningful ultraviolet, flavor, dynamical, or boundary-condition theories that enlarge the comparison class. The theorem is therefore a classification-rigidity result: it closes the declared same-scope residual problem, not the space of all conceivable enlarged-domain physics. Terminality here concerns admissible residual classification within the fixed representation regime, not completeness of physical ontology or finality of future physical theory.*

Proposition 2.4 (Compatibility with enlarged-domain physics). *The present theorem is not anti-BSM, anti-UV, anti-flavor, or anti-boundary physics. If an enlarged-domain theory T^+ derives N_g or a residual value by adding anchors, tensor witnesses, transport structure, boundary data, dynamical laws, flavor symmetries, matching conditions, or ultraviolet completion data, then T^+ is not a counterexample to the fixed-scope theorem. It is a new scope whose projection may instantiate or explain the old fixed-boundary data.*

Proof. A derivation in T^+ either preserves the old residual quotient or it does not. If it preserves the old quotient and adds no new witness authority, it factors through the same terminal status map and gives no new same-scope tightening. If it derives a value by using additional structure, then at least one component of S_{res} has changed: witness package, anchor set, transport family, standing quotient, boundary ledger, or admissible operator family. That is enlarged-domain physics. Such physics is compatible with the fixed-scope theorem precisely because the theorem classifies only operations preserving S_{res} . $\square$

Remark 2.5 (Why this is not a retreat). *Certifying generation count and residual numerical values as externally fixed is not failure to analyze them. It is the correct fixed-scope classification. The theorem determines what kind of object these residuals are: not skin, not same-interior derivable tensor content, not hidden selector output, but fixed-boundary data whose roles are structurally fixed and whose values require boundary matching or a larger-domain witness.*

Remark 2.6 (Classification legitimacy: global form). *The classifications used here are not semantic relabelings. They are exhaustion partitions induced by admissibility-preserving equivalence, witness preservation, and same-scope residual closure. A residual receives a status only after the relevant role, witness, quotient, repair, and boundary tests have been applied. The mathematical work done by the classification is therefore exclusionary: it removes proposed same-scope alternatives by showing that they either identify away, are already forced, are pure skin, are inadmissible selectors, or leave the declared comparison class.*

Remark 2.7 (Distinction from ordinary underdetermination). *Ordinary underdetermination leaves multiple admissible alternatives open inside a comparison class. The present result has a different form. It partitions the residual candidates by terminal status and proves that each audited route either factors through the terminal residual quotient, collapses to skin or identification, is already witness-forced, imports inadmissible selector authority, or changes scope. Thus the result is not merely that several parameter choices remain empirically possible; it is that the same-scope residual problem has an exhaustive structural classification.*

3 Relation to upstream Standard Model closure

The argument is downstream of prior Standard Model representation results. It assumes that the admissible representation skeleton, representation-family coexistence, charge-lattice rigidity, and parameter-normal-form discipline have already been fixed. The purpose here is not to repeat those derivations, but to classify the remainders they leave inside the same comparison regime.

Upstream source role	Imported closure function	Use in this paper
SM-FINAL	Fixes the Standard Model representation envelope and the admissible occupant/load class.	Supplies K_{SM} and $\mathcal{E}_{\text{SM}}$.
SM-PT-1–SM-PT-3	Close prior parameter-tightening stages and representation-family coexistence.	Defines the residual candidate space audited here.
Standard Model classification theorem	Distinguishes occupant uniqueness from irreducible fixed-boundary inputs.	Prevents unsupported strengthening: numerical values can be externally fixed without creating a second occupant.
Structura Ex Necessitate Standardis Modelis	Fixed-load interior closure.	Blocks same-interior escape routes that would change the declared SM load while claiming same scope.

Charge-lattice rigidity	Fixed-scope charge labels factor through the same discrete lattice up to normalization/sign.	Prevents exotic charge assignments from masquerading as residual parameter freedom.
Gauge-coupling parameter closure	Raw chart-numeric coupling values are not standing-bearing as raw numbers.	Distinguishes parameter slots and invariant relations from arbitrary chart labels.
Anchor/Tensor/Skin decomposition	Every descriptive element is anchor, tensor, or skin.	Supplies the role test for residual candidates.
AMetric boundary	No selector, ranking, metric, typicality, or canonical representative can act as boundary authority in fixed scope.	Blocks generation and parameter selectors.
Same-scope operator exhaustion	Every same-scope operator is representation-equivalent or inadmissible.	Blocks hidden totalization and future-binder routes.
Standing-admissibility identity closure	No second same-domain classifier can govern the old quotient.	Blocks a separate generation/value classifier.

Remark 3.1 (Interior closure versus residual closure). *Interior closure proves that the declared Standard Model representation structure is fixed under same-load admissible operations. Residual closure proves that the remaining degrees of freedom cannot be further tightened by same-scope operations. These are different theorems. The first fixes the occupant; the second classifies what remains inside or at the boundary of that occupant.*

4 Local AASC machinery

The tests MR, RPC, and GCI are reconstructed locally and supplemented by the role and boundary tests needed for a self-contained residual classification.

Definition 4.1 (Maximality Remainder test). *The Maximality Remainder test MR asks whether a residual candidate X leaves an admissibility-bearing distinction unclassified inside the fixed scope. It has three outcomes:*

- (i) *MR-binding exclusion: X expands the admissible representation space without a declared witness or coherence necessity.*
- (ii) *MR-binding identification: X changes no standing-bearing witness and is therefore quotient-identical to an already admitted representative.*
- (iii) *MR-non-binding structural deferral: X is standing-bearing but no same-scope witness, transport, or coherence condition binds its value; it is externally fixed.*

Definition 4.2 (Repair-Prohibition Check). *The Repair-Prohibition Check RPC rejects any attempt to rescue an otherwise inadmissible residual by later filtering, empirical back-import, auxiliary restriction, hidden matching, privileged chart selection, or post-hoc tuning while continuing to claim same-scope derivation.*

Definition 4.3 (Global Coherence Intersection). *The Global Coherence Intersection GCI is the common witness intersection of all declared same-scope coherence constraints. A non-unity multiplicity or nontrivial residual relation is retained only when it is forced by every relevant standing-preserving coherence route in W_{res} . If a proposed multiplicity is not forced by GCI, it is either redundant skin, discretionary restriction, or scope change.*

Definition 4.4 (Anchor/Tensor/Skin role test). *The role test ATS classifies a residual element x as follows:*

- (i) *x is an anchor if moving it changes definedness, boundary typing, domain identity, or envelope membership.*
- (ii) *x is a tensor if moving it under fixed anchors changes quotient-level invariant content, witness compatibility, or admissible continuation.*
- (iii) *x is skin if moving it changes no standing/admissibility classification and no coherent retained-trajectory structure.*

No fourth same-scope role exists.

Definition 4.5 (AMetric Boundary Test). *The AMetric Boundary Test ABT rejects any same-scope boundary-authority attempt using a selector, order, rank, metric, probability, typicality, threshold, proximity relation, canonical representative, gate chooser, or privileged presentation to bind a residual that is not already bound by the declared witness package.*

Definition 4.6 (Standing-Gate Closure). *Standing-Gate Closure SGC states that admissibility and standing cannot be governed by a second same-domain classifier once the standing quotient is fixed. A new classifier is either extensionally identical to the old gate, skin, or a scope-changing/inadmissible boundary import.*

5 Residual candidate space

Let $\mathcal{X}_{\text{res}}$ be the set of admissibility-bearing residual candidates after SM-PT-3. The residual analysis separates residual candidates from rival routes. A residual candidate is an object still typed inside S_{res} ; a rival route is an attempted operation that may fail before entering S_{res} .

Definition 5.1 (Residual candidates). *The same-scope residual candidate space contains the following classes:*

- R1. M_{∞} : unbounded repetition of an admitted structure.*

R2. M_b : bounded but unforced repetition of an admitted structure.

R3. M_c : bounded multiplicity witness-required by GCI.

R4. N_g : generation cardinality as declared by the fixed envelope.

R5. Θ_{res} : residual standing-bearing numerical parameter slots not internally fixed by prior closure.

R6. S_{skin} : presentation, convention, coordinate, basis, normalization, or chart residue that changes no witness.

Definition 5.2 (Rival routes). *The following are not successful residual candidates merely because they are proposed:*

Q1. hidden generation or value selector;

Q2. empirical value promoted to internal derivation;

Q3. metric, typicality, anthropic, probabilistic, optimization, or naturalness authority;

Q4. repair of a failed or previously unbound derivation;

Q5. added BSM or UV sector presented as same-scope;

Q6. hidden label or latent variable with no admitted witness;

Q7. partial operator totalized on the old carrier.

Each route is audited in section 15.

Definition 5.3 (Terminal status map). *A terminal status map is a partial function*

$$\tau : \mathcal{X}_{\text{res}} \rightarrow \{\text{EXCLUDE, IDENTIFY, FORCE, Skin, EXT-FIX}\}$$

defined exactly for same-scope residual candidates. Rival routes that import selectors, change scope, or repair a failed derivation are not assigned a sixth residual status; they fail the domain condition for τ .

6 The no-sixth-status theorem

The paper's central result is the following status exhaustion theorem.

Theorem 6.1 (Residual status exhaustion). *Every same-scope residual candidate $X \in \mathcal{X}_{\text{res}}$ has exactly one terminal status:*

$$\tau(X) \in \{\text{EXCLUDE, IDENTIFY, FORCE, Skin, EXT-FIX}\}.$$

No sixth same-scope terminal status exists.

Proof. Let $X \in \mathcal{X}_{\text{res}}$. Apply the role test. If varying X changes definedness, boundary typing, or envelope membership, then X is anchor-level. Inside the fixed scope the anchor cannot be moved. If the anchor-level datum is required for instantiation but not internally fixed, it is externally fixed; if a proposed motion tries to keep the old scope, it is not a same-scope residual.

Suppose definedness is fixed. If varying X changes a declared quotient-level witness, compatibility relation, transport relation, or coherent continuation fact, then X is tensor-level. If the declared witness package mechanically fixes the residual, $\tau(X) = \text{FORCE}$. If the tensor variation expands the old representation space without a declared coherence witness, MR binds exclusion, so $\tau(X) = \text{EXCLUDE}$. If the tensor-level slot is real but no same-scope witness binds its value, MR is structurally non-binding and the slot is externally fixed, so $\tau(X) = \text{EXT-FIX}$.

Suppose varying X changes no definedness condition and no tensor witness. Then X is skin and $\tau(X) = \text{Skin}$, or it is identified with an already admitted representative, $\tau(X) = \text{IDENTIFY}$, depending on whether the residue is retained as presentation or quotient-collapsed as redundancy. The distinction is representational only; neither case yields a new standing-bearing residual.

A sixth status would need to change constructional behavior while changing neither anchor nor tensor nor skin, or it would need to bind a residual while preserving all old witnesses and without importing any selector. The first option contradicts the role exhaustion theorem. The second option contradicts the AMetric no-selector/no-parameter boundary and same-scope operator exhaustion. Therefore no sixth same-scope status exists. $\square$

Corollary 6.2 (No internal-but-not-boundary remainder). *There is no residual class whose status is “not derivable internally, but also not externally fixed,” inside S_{res} . Once skin and quotient-identification are removed, every standing-bearing unbound residual is structurally externally fixed.*

Remark 6.3 (Classification legitimacy). *The classification in theorem 6.1 is not a semantic relabeling of ignorance. It is an exhaustion partition induced by admissibility-preserving equivalence, the anchor/tensor/skin role test, same-scope witness completion, and the no-selector/no-repair discipline. A proposed residual distinction has mathematical force only if it changes a declared witness, transport relation, standing quotient, boundary record, or coherent continuation. Otherwise it is skin, representation, or inadmissible selector authority. Thus the theorem classifies operational roles under the declared comparison class rather than merely naming unresolved physics.*

7 Witness completion and competitor discipline

A hardened closure theorem must state what a counterexample would have to preserve.

Definition 7.1 (Residual witness completion). *The residual witness package W_{res} is witness-complete when every same-scope residual distinction affecting admission, standing-bearing reuse, quotient identity, admissible transport, boundary matching, or coherent continuation is detected by at least one declared witness in W_{res} .*

Lemma 7.2 (No arbitrary witness import). *A proposed additional witness w' cannot be added to W_{res} while preserving the same residual scope unless w' factors through the old witness quotient. If w' changes a residual status, the witness package has changed and the proposal is an enlarged scope. If w' changes no residual status, it is representation.*

Proof. If w' affects terminal classification, then at least one of the old comparison components changes: witness family, quotient relation, transport behavior, envelope boundary, or standing gate. By the same-scope competitor definition, the proposal is not same-scope. If none changes, then w' adds no residual classification authority and factors through the old quotient as representation-equivalent. A third possibility would require w' to act as an independent same-domain classifier, which is blocked by SGC. $\square$

Theorem 7.3 (Fixed-witness residual closure). *Inside a witness-complete residual scope, every same-scope residual competitor preserving W_{res} either induces the same terminal status map τ , collapses to skin/identification, or fails admissibility. A competitor that changes τ changes the witness package or the envelope.*

Proof. Let P be a same-scope competitor preserving W_{res} . Because it preserves the witness package, it preserves all declared locations at which a residual can affect standing. By role exhaustion, any difference introduced by P is anchor, tensor, or skin. Anchor motion changes envelope membership and is not same-scope. Tensor motion is detected by W_{res} ; since P preserves W_{res} , it cannot alter the tensor status without contradiction. Skin motion is quotient-inert. Therefore P induces the same τ , except for purely notational changes. If P attempts to select a different residual status without witness change, it imports a selector or second classifier and fails admissibility. $\square$

8 Multiplicity exhaustion

Multiplicity is the easiest residual to overstate and the easiest to audit incorrectly. The proof distinguishes unbounded repetition, bounded unforced repetition, coherence-forced multiplicity, gauge/presentation multiplicity, and envelope cardinality. They are not the same object.

Definition 8.1 (Multiplicity classes). *A multiplicity claim inside S_{res} has one of four forms:*

- (i) *unbounded multiplicity: arbitrary repetition of an admitted structure with no terminal bound;*
- (ii) *bounded unforced multiplicity: finite repetition whose bound is not required by W_{res} ;*
- (iii) *coherence-forced multiplicity: finite repetition required by GCI;*
- (iv) *presentation multiplicity: multiple names, charts, bases, or representatives for the same standing class.*

Generation cardinality is treated separately in section 9 because it is an envelope cardinality in the fixed Standard Model scope, not a discretionary same-interior repetition operator.

Lemma 8.2 (Illicit unbounded multiplicity). *Unbounded repetition of an admitted same-scope structure is excluded.*

Proof. Unbounded repetition strictly enlarges the admissible representation space while preserving no declared witness that requires the enlargement. If the added copies change no witness, they are redundant and collapse under quotienting; if they change a witness, the witness package and envelope have changed; if they are retained without witness change, they are hidden labels. Thus MR binds exclusion and RPC blocks post-hoc repair by auxiliary restriction. $\square$

Corollary 8.3 (Terminal status of unbounded multiplicity). *For M_∞ , $\tau(M_\infty) = \text{EXCLUDE}$.*

Lemma 8.4 (Eliminative collapse of bounded unforced multiplicity). *A bounded multiplicity not required by global coherence is identified away or excluded; it cannot remain as a discretionary standing-bearing multiplicity.*

Proof. Let M_b be a finite multiplicity bound not forced by GCI. If the multiple copies produce no standing-bearing distinction, the standing quotient identifies them; retaining the number as structure would treat skin as tensor. If the copies produce a standing-bearing distinction, then the distinction is not licensed by the declared witness package and changes scope. If a rule selects the finite bound while no witness requires it, the rule is a selector. These alternatives exhaust the case. Hence the admissible same-scope disposition is identification when the multiplicity is witness-inert and exclusion when it is witness-expanding without declared coherence force. $\square$

Proposition 8.5 (Coherence-forced multiplicity). *A nontrivial bounded multiplicity may survive only when witness-forced by GCI.*

Proof. If GCI requires a non-unity multiplicity, then the multiplicity is not discretionary. It is a tensor-level compatibility requirement of the declared witness package. In that case $\tau(M_c) = \text{FORCE}$. If the requirement is absent, the preceding lemma applies. $\square$

Remark 8.6 (Why this does not collapse generation count). *The collapse of bounded unforced multiplicity applies to same-interior repetition of an already admitted structure. It does not authorize changing a declared envelope cardinality while claiming the old envelope. The generation count in the empirical Standard Model representation regime is part of the envelope ledger audited in section 9. Collapsing it to unity would move the envelope, not tighten a same-scope residual.*

9 Generation count as externally fixed cardinality

The generation-count question is the central residual target. The precise statement must be precise: this paper does not derive $N_g = 3$ internally. It proves that the generation cardinality cannot be bound by any admissible same-scope residual-tightening operation after the declared Standard Model envelope is fixed.

Definition 9.1 (Generation cardinality). *The generation cardinality N_g is the envelope-level cardinality of repeated fermion family roles admitted by the fixed Standard Model representation regime. It records how many copies of the already typed representation-family profile instantiate the envelope. In the empirical Standard Model envelope this value is three; the present theorem classifies the status of that value rather than deriving it.*

Proposition 9.2 (Generation count is not a presentation coordinate). *The generation cardinality is not skin. Moving N_g changes the envelope ledger: flavor slots, mass/Yukawa dimensions, mixing-coordinate dimensions, family-indexed data, and boundary matching records.*

Proof. A skin coordinate can be changed without altering standing-bearing witnesses or coherent continuation. Changing N_g changes the number of field-role instances and the dimensionality of the associated parameter slots and flavor witnesses. Even if some anomaly relations replicate generationwise, the instantiated representation object and parameter ledger are not the same object. Thus N_g is not a mere chart, basis, or naming convention. $\square$

Proposition 9.3 (No same-scope generation selector). *No same-scope admissible operation selects a unique numerical generation cardinality inside S_{res} unless that cardinality is already declared in the envelope or forced by an admitted witness.*

Proof. Let B_N be a proposed generation selector. If B_N depends on a metric, probability, typicality, naturalness ranking, anthropic filter, optimization rule, canonical representative, or privileged presentation, it fails ABT. If B_N changes flavor witnesses, field-role membership, mixing dimensions, or boundary matching records, it changes the envelope. If it changes none of these, it cannot distinguish one value of N_g from another as standing-bearing. If it claims authority anyway, it is a second same-domain classifier, blocked by SGC. Therefore no same-scope selector exists. $\square$

Lemma 9.4 (No hidden generation binder). *A hidden internal binder for N_g either factors through the old witness quotient, changes scope, or is an inadmissible selector.*

Proof. Suppose H_N is hidden. If it preserves all old witnesses and changes no quotient or envelope record, then it has no standing-bearing effect and is skin or representation-equivalent status. If it changes the terminal classification of N_g , then it changes at least one witness, transport, role, or envelope component. That is a scope change. If it is used to select a value while neither changing nor factoring through the old witnesses, then it is precisely a selector without admitted authority. The same-scope operator-exhaustion theorem closes attempts to totalize H_N by composition, iteration, fixed-point detour, or future definitional extension. $\square$

Proposition 9.5 (Generation-lifting exhaustion). *Every admissible same-scope route that purports to force generation cardinality reduces to one of the following cases:*

- (i) *anomaly transport: the anomaly/witness ledger is already preserved generationwise or by the declared family package; if it selects a new cardinality, the witness package has changed;*

- (ii) *representation lifting: repeated representation-family profiles are either witness-inert copies, coherence-forced tensor content, or envelope-load changes;*
- (iii) *flavor redundancy: a family label that changes no flavor, Yukawa, mixing, or boundary record is skin/identification, while a label that changes any such record changes the envelope ledger;*
- (iv) *hidden quotient multiplicity: quotienting family copies without changing witnesses is representation; quotienting them while changing value slots or mixing dimensions is scope change;*
- (v) *transport-generated family lifting: admissible transport preserves the old family-role quotient; if transport creates a new cardinality constraint, the transport family has been enriched;*
- (vi) *selector insertion: a rule choosing a cardinality without witness force fails the AMetric boundary and standing-gate closure;*
- (vii) *envelope enlargement: a UV, flavor, sterile-sector, boundary, or BSM mechanism may select a cardinality only by adding anchors or tensor witnesses and hence by leaving the fixed residual scope.*

No listed route yields an inequivalent same-scope derivation of N_g .

Proof. A proposed route to N_g must either use already declared witnesses, introduce a new witness, act only on representation/presentation labels, alter transport, or add external/boundary structure. Already declared witnesses either force the value, contrary to the residual assumption, or leave it externally fixed. New witnesses change scope. Presentation labels are skin or quotient-identifications. Transport that generates new family cardinality authority is enriched transport, not the old T_{SM} . External or boundary structure is an envelope enlargement. The only remaining option is a selector acting without witness force, which is blocked by ABT and SGC. Hence the route list is exhaustive. $\square$

Theorem 9.6 (Generation reconstruction equivalence). *Any admissible same-scope reconstruction of generation cardinality from hidden quotient structure, sectorization, internal topology, flavor-sector stratification, or dynamical partitioning is equivalent to one of three cases:*

- (i) *multiplicity encoding already exhausted by the multiplicity and generation-lifting results;*
- (ii) *hidden selector structure acting without witness authority;*
- (iii) *enlarged-regime structure that changes the skeleton, witness package, transport family, standing quotient, or boundary ledger.*

Therefore no emergent-topology or sectorization route yields an inequivalent same-scope derivation of N_g .

Proof. Let R_N be such a reconstruction. If its quotient, topology, sector, or dynamical partition changes no declared witness, no transport behavior, and no boundary record, then it merely recodes the already admitted family multiplicity and is handled by the multiplicity-collapse and generation-lifting analyses. If it distinguishes one cardinality from another while preserving all such witnesses, it selects among witness-equivalent residual possibilities and is a hidden selector, blocked by ABT and SGC. If it succeeds because the quotient, sector, topology, or dynamics supplies new invariant structure, then the witness package or admissible transport family has been enriched; the proposal is enlarged-regime physics, not same-scope residual tightening. These cases exhaust hidden reconstruction routes. $\square$

Remark 9.7 (Dualization and sector-replication collapse). *Same-scope family duplication by admissible dualization, mirror sectorization, conjugate-sector representation, internal sector replication, or representation-dual representation does not evade the generation analysis. If the duplication changes no declared witness, transport behavior, mixing dimension, family-indexed parameter slot, or boundary record, it is multiplicity encoding, skin, or quotient identification. If it changes any of those structures, it adds witness content or changes the envelope. Thus dualization-based family production collapses to theorems 8.4, 9.5 and 9.6.*

Theorem 9.8 (Generation-count fixed-boundary theorem). *Within the fixed Standard Model residual scope S_{res} , generation cardinality is structurally externally fixed:*

$$\tau(N_g) = \text{EXT-FIX}.$$

No admissible same-scope residual-tightening operation derives, selects, collapses, or changes N_g without changing the envelope, importing a selector, or violating standing preservation.

Proof. By the preceding non-skin proposition, N_g is not a chart, basis, or naming coordinate. By the no-selector proposition and hidden-binder lemma, no same-scope internal operation binds its value. By theorems 9.5 and 9.6, the standard rival routes—*anomaly transport, representation lifting, flavor redundancy, hidden quotient multiplicity, transport-generated family lifting, selector insertion, and envelope enlargement*—either factor through the old quotient, fail admissibility, or leave the fixed scope. By definition, changing N_g changes the envelope ledger and associated parameter/mixing structures. Therefore N_g is an anchor/fixed-boundary datum required to instantiate the fixed Standard Model representation regime. Since it is not internally bound by the same-scope witness package, MR is structurally non-binding and $\tau(N_g) = \text{EXT-FIX}$. $\square$

Corollary 9.9 (Correct reading of “not derived”). *The statement that N_g is not derived here does not mean that all generation counts are same-scope alternatives. It means that the fixed Standard Model theorem classifies the observed generation cardinality as a fixed-boundary datum. A larger-domain theory may attempt to derive it, but that derivation is not a same-scope tightening of the fixed residual theorem.*

10 Residual parameter exhaustion

Residual numerical parameters require a parallel but not identical analysis. Some parameters are pure skin; some are forced relations; some are standing-bearing but not same-scope derivable values. The target is to classify them without confusing raw chart labels with invariant content.

Definition 10.1 (Residual parameter vector). *Let Θ_{res} denote the collection of numerical or functional slots not fixed by prior same-scope closure. It may include, depending on the declared envelope:*

- (i) *gauge coupling value slots and running-coupling chart coordinates;*
- (ii) *Yukawa eigenvalue or mass parameter slots;*
- (iii) *mixing angles and invariant phase residues after quotienting basis conventions;*
- (iv) *Higgs-sector values and scale-setting or matching parameters;*
- (v) *CP-like or boundary-condition residues;*
- (vi) *normalization, scheme, chart, gauge, or basis coordinates.*

Definition 10.2 (Parameter role partition). *The residual parameter vector decomposes as*

$$\Theta_{\text{res}} = \Theta_{\text{skin}} \sqcup \Theta_{\text{force}} \sqcup \Theta_{\text{env}},$$

where Θ_{skin} contains chart, basis, convention, gauge, or redundant coordinates; Θ_{force} contains parameters or relations mechanically fixed by the declared witness package; and Θ_{env} contains standing-bearing value slots required to instantiate the envelope but not internally fixed by same-scope admissibility.

Lemma 10.3 (Skin parameter collapse). *Any raw parameter coordinate whose value changes under admissible redescription while all tensor witnesses remain fixed is skin. It cannot be promoted to standing-bearing residual content as a raw number.*

Proof. If a coordinate changes under an admissible chart, scheme, basis, or normalization redescription while the same physical/witness content is preserved, then treating the coordinate value as tensor identity would assign standing to presentation. That violates role discipline. The admissible status is Skin or quotient-identification, not a new parameter residue. $\square$

Lemma 10.4 (Forced-parameter retention). *If a residual relation or value is witness-required by the declared witness package, it is not a residual freedom. It belongs to Θ_{force} and has terminal status FORCE.*

Proof. A value or relation required by W_{res} under fixed anchors is tensor-level coherence content. It is retained because moving it changes witness compatibility. Such a parameter is not discretionary and does not require external fixation. $\square$

Proposition 10.5 (No selective same-scope parameter binding). *Let $\theta \in \Theta_{\text{env}}$. No admissible same-scope operation can bind θ to a unique numerical value while leaving the fixed residual scope unchanged.*

Proof. Let B_θ be a proposed binder. If B_θ uses empirical input, boundary matching, or a larger-domain relation, then it is an boundary import or scope extension, not a same-scope derivation. If it uses a metric, typicality rule, naturalness measure, optimization principle, canonical chart, or privileged representative, it fails ABT. If it changes a tensor witness, it changes the residual witness package. If it changes no witness, then it cannot determine standing-bearing numerical content; at most it picks a representative. If it binds only one parameter while the same witness logic leaves analogous parameters unbound, it introduces a selective classifier. SGC and same-scope operator exhaustion block that classifier. Hence no same-scope binder exists. $\square$

Definition 10.6 (Collective parameter binder). *A collective parameter binder is a same-scope operation B_Θ that does not necessarily select one parameter in isolation but instead purports to restrict a nontrivial relation, submanifold, fixed point, hierarchy, or coupled value assignment among a subset of Θ_{env} .*

Theorem 10.7 (Collective parameter non-closure). *No admissible same-scope collective parameter binder can nontrivially restrict Θ_{env} unless the restriction is already an element of Θ_{force} . Any new collective restriction on externally fixed parameters is one of the following: skin/coordinate choice, already-forced witness relation, hidden selector, empirical or boundary import, repair, or scope change.*

Proof. Let B_Θ purport to restrict a collective parameter family. If the restriction is invariantly required by the declared witness package, then it was not in Θ_{env} ; it belongs to Θ_{force} . If the restriction depends on a basis, scheme, normalization, or chart, it is skin or quotient representation. If it uses renormalization trajectories, fixed-point language, matching scales, flavor manifolds, or ultraviolet relations to pick a value or submanifold, then the value selection requires boundary data or additional tensor witnesses; this is boundary import or scope change. If it picks a submanifold without such witnesses, it distinguishes witness-equivalent parameter points and is therefore a selector or second classifier. If it is introduced only after an unbound parameter is discovered, it is repair. Thus collective binding does not evade the selective-binding obstruction. $\square$

Corollary 10.8 (No coupled-manifold loophole). *The failure of a same-scope binder for individual Yukawa values, mixing coordinates, coupling values, CP residues, or scale anchors cannot be bypassed by asserting an unlicensed coupled parameter manifold. A coupled manifold is admissible only when it is already witness-forced, pure skin, or supplied by enlarged-domain/boundary structure.*

Remark 10.9 (Renormalization and fixed-point compatibility). *Renormalization-group flow, fixed-point structure, matching-scale arguments, and ultraviolet boundary conditions are compatible with the present theorem. They do not evade it unless they determine residual values inside the*

same witness package and boundary ledger without adding selector authority. By residual quotient terminality and factorization (theorems 13.6 and 13.11), any RG or fixed-point procedure that preserves the fixed witness fiber factors through the same residual quotient; it therefore cannot create a new value-classifier. If it determines values by supplying flow endpoints, scale anchors, matching conditions, stability criteria, or ultraviolet data, it has added boundary input or enlarged-regime structure. If it supplies no such additional data, it defines trajectories or coordinate descriptions but does not internally force the residual value.

Theorem 10.10 (Residual parameter exhaustion). *Every residual parameter in Θ_{res} is either skin/identified, coherence-forced, or externally fixed:*

$$\Theta_{\text{res}} = \Theta_{\text{skin}} \sqcup \Theta_{\text{force}} \sqcup \Theta_{\text{env}},$$

$$\tau(\Theta_{\text{skin}}) \in \{\text{Skin}, \text{IDENTIFY}\}, \quad \tau(\Theta_{\text{force}}) = \text{FORCE}, \quad \tau(\Theta_{\text{env}}) = \text{EXT-FIX}.$$

No fourth parameter class survives inside the fixed residual scope.

Proof. Apply ATS to each parameter slot. If variation changes no witness, the parameter is skin or identified. If variation changes a declared witness and the witness package fixes the value or relation, the parameter belongs to Θ_{force} . If variation changes standing-bearing instantiation data but the witness package does not bind a unique value, then the slot is externally fixed. A fourth status would require a parameter to be standing-bearing, unforced, not skin, and not externally fixed. Binding such a parameter singly would require a selector, second classifier, repair, empirical back-import, or scope change. Binding such parameters collectively is blocked by collective parameter non-closure unless the relation was already forced. Therefore the displayed partition is exhaustive. $\square$

Corollary 10.11 (Irreducible inputs do not multiply the occupant). *Different assignments to externally-fixed residual datum parameters instantiate different parameter points or boundary-matched models inside the same representation structure. They do not create distinct same-scope occupants unless the assignment changes the witness family, standing quotient, role typing, transport structure, or envelope boundary.*

11 Externally fixed residual data do not mean arbitrary

A common objection is that classification as externally fixed simply hides ignorance. That objection confuses value derivation with role classification.

Definition 11.1 (Externally fixed residual datum). *Operationally, a residual datum is externally fixed when its admissible role is structurally determined inside S_{res} while its numerical, cardinal, or specifying value is not internally forced by any admissible same-scope operation. The adjective “external” is relative to the fixed residual regime only: it means fixed at the boundary ledger or by an enlarged regime, not metaphysically outside physics. Equivalently, x is externally fixed when all of the following hold:*

- (i) x is required to instantiate the declared representation envelope;
- (ii) x is not pure skin and does not identify away under admissible redescription;
- (iii) the same-scope witness package does not mechanically bind its value;
- (iv) every operation that would bind its value imports a selector, changes scope, or supplies new boundary/larger-domain data.

Thus an externally fixed residual datum differs from an ordinary unconstrained knob: the theorem fixes its role, slot, quotient behavior, admissible transformation discipline, and boundary location, while refusing only an illicit same-scope derivation of its value.

Proposition 11.2 (Externally fixed residues are structurally fixed). *If x is externally fixed, then its value is not internally derived by the same-scope theorem, but its role, slot, quotient behavior, admissible transformation law, and boundary location are fixed by the theorem.*

Proof. Externally fixed status is assigned only after skin, identification, forced-coherence, and exclusion routes are exhausted. Thus the datum is not arbitrary presentation. It is a fixed role in the envelope ledger. What remains unbound is the numerical/cardinal value, not the structural status of the datum. $\square$

Definition 11.3 (Fixed-boundary irreducible residual). *A residual x is fixed-boundary irreducible in S_{res} when x is externally fixed and every purported same-scope derivation of its value requires at least one of the following: hidden selector structure, illicit witness enlargement, envelope transport, empirical back-import, repair, or non-standing-preserving totalization.*

Theorem 11.4 (Fixed-boundary irreducibility). *Every standing-bearing unbound residual that survives residual status exhaustion is fixed-boundary irreducible inside the fixed Standard Model residual scope. In particular, N_g and every $\theta \in \Theta_{\text{env}}$ are not merely “not yet derived”; any same-scope derivation of their values would have to leave the admissible fixed-scope comparison class.*

Proof. Let x be a standing-bearing residual surviving status exhaustion. It is not excluded, identified, forced, or skin. By the no-sixth-status theorem, x is externally fixed. Suppose a same-scope derivation D_x of its value exists. If D_x preserves the witness package and quotient while selecting among witness-equivalent values, it is a selector or second classifier, contrary to ABT and SGC. If D_x avoids selector status by adding a witness, anchor, boundary condition, transport law, or larger-domain relation, then D_x changes S_{res} . If D_x relies on empirical calibration, after-the-fact filtering, or tuning, it is boundary import or repair. If D_x is obtained by extending a partial operation to a total one, same-scope operator exhaustion classifies the extension as representation-equivalent or non-admissible. Therefore every derivation route leaves fixed-scope admissibility, proving fixed-boundary irreducibility. $\square$

Corollary 11.5 (Externally fixed is not provisional ignorance). *Within S_{res} , a fixed-boundary irreducible residual is not a placeholder awaiting an unspecified internal trick. A successful derivation must either already factor through the declared witness package or introduce new admissible structure, in which case it is an enlarged-scope derivation.*

Corollary 11.6 (Boundary and larger-domain compatibility). *A larger-domain construction may attempt to derive an externally fixed datum. If it succeeds, it has supplied additional anchors or tensor witnesses. Its result may project back to the old envelope, but the derivation itself belongs to the enlarged scope.*

12 Hidden selector and totalization obstruction

A central structural layer is the hidden-selector obstruction. It prevents the claim that an unspecified future internal principle could bind generation count or residual parameter values while preserving the old scope.

Definition 12.1 (Admissible internal binder). *An admissible internal binder for a residual x is a same-scope operation B_x such that:*

- (i) B_x preserves S_{res} ;
- (ii) B_x preserves the standing quotient and witness package;
- (iii) B_x introduces no selector, metric, typicality rule, empirical back-import, repair, or scope extension;
- (iv) B_x determines a unique terminal value or status for x not already determined by W_{res} .

Theorem 12.2 (No hidden residual binder). *There is no admissible internal binder for N_g or for any $\theta \in \Theta_{\text{env}}$.*

Proof. Assume B_x exists. If B_x determines a value not already determined by W_{res} while preserving W_{res} , then it distinguishes between witness-equivalent possibilities. That is a selector or second classifier, contrary to ABT and SGC. If B_x avoids selector status by adding a new witness, then it no longer preserves S_{res} . If it changes no witness and no classification fact, then it does not bind x . If it binds x by extending a partial operation on the old carrier, then same-scope operator exhaustion classifies the move as domain enlargement, carrier collapse, evaluative completion, conservative representation equivalence, or non-conservative family change; none is a nontrivial same-scope binder. Contradiction. $\square$

Corollary 12.3 (No future-principle loophole inside fixed scope). *A future principle can bind N_g or a residual parameter only if it supplies new admissible witnesses or boundary data. In that case it is valuable larger-domain physics, not a same-scope completion of the fixed residual theorem.*

Definition 12.4 (Deferred internalization route). *A deferred internalization route is a claim that a principle not presently named by the fixed residual theorem will later bind N_g or Θ_{env} while preserving the same residual scope, witness package, standing quotient, transport family, and envelope boundary.*

Theorem 12.5 (No deferred internalization). *No deferred internalization route exists inside S_{res} . A future principle that is genuinely same-scope is already an admissible operation on the present residual quotient and is exhausted by the same-scope operator closure. A future principle that is not already such an operation supplies new witnesses, anchors, transport, boundary data, or selector authority and is therefore enlarged-domain rather than same-scope.*

Proof. Let F be a proposed future principle. If F preserves S_{res} , then by definition it acts on the same residual candidates, the same witness quotient, and the same terminal-status map. It is therefore an admissible same-scope operator and is classified by same-scope operator exhaustion as representation-equivalent, skin, already forced, or inadmissible as a binder. If F does not preserve S_{res} , then it changes at least one of the structures that define the fixed comparison class. That may be a legitimate new physical theory, but it is not deferred internal completion of the fixed residual theorem. Thus the future-principle loophole is closed without forbidding enlarged-domain work. $\square$

13 Residual reconstruction category

The uniqueness statement is a quotient statement, not a claim that every notation or presentation is literally identical. This section makes the closure claim functorial enough to block a parallel same-scope classifier.

Throughout this section, $\mathbf{C}_{\text{res}}$ denotes the fixed-witness residual reconstruction category, $\mathbf{Q}_{\text{res}}$ denotes its admissible residual quotient, $\pi_{\text{res}} : \mathbf{C}_{\text{res}} \rightarrow \mathbf{Q}_{\text{res}}$ denotes the quotient functor, Q_{τ} denotes the object-level terminal status quotient, and $q_{\text{res}} : \mathcal{X}_{\text{res}} \rightarrow Q_{\tau}$ denotes the candidate-level quotient map. Morphisms are always required to preserve the declared same-scope data unless explicitly marked as enlarged-domain.

Definition 13.1 (Residual reconstruction object). *Let $\mathbf{C}_{\text{res}}$ be the category whose objects are finite residual reconstruction systems*

$$C = (S_{\text{res}}, \mathcal{X}, \tau, W_{\text{res}}, T_{\text{SM}}, \sim_{\text{st}}, \mathcal{E}_{\text{SM}}),$$

where $\mathcal{X}$ is a residual candidate family and τ is a terminal status map.

Definition 13.2 (Admissible residual morphism). *A morphism $\Phi : C \rightarrow C'$ in $\mathbf{C}_{\text{res}}$ is admissible when it satisfies:*

- M1. skeleton preservation: it preserves field-role, representation-family, charge-lattice, and interaction slots;*
- M2. witness preservation: it preserves W_{res} and the induced standing quotient;*

- M3. *transport preservation: it commutes with T_{SM} up to admissible redescription;*
- M4. *terminal-status preservation: it sends excluded, identified, forced, skin, and externally fixed residues to the corresponding quotient classes;*
- M5. *boundary covariance: it may relabel boundary records but may not move externally fixed data into same-interior derivations without new scope declaration;*
- M6. *no-selector/no-repair discipline: it introduces no hidden selector, repair, empirical back-import, or boundary authority.*

Proposition 13.3 ($\mathbf{C}_{\text{res}}$ is a category). *Residual reconstruction objects with admissible residual morphisms form a category.*

Proof. Identity maps preserve all listed structures. Compositions of skeleton-preserving, witness-preserving, transport-preserving, terminal-status-preserving maps preserve the same structures by transitivity. A selector or repair cannot appear in a composite unless it appears in one factor, which admissibility forbids. Associativity is inherited from ordinary composition. $\square$

Definition 13.4 (Admissible same-scope residual classifier). *An admissible same-scope residual classifier is a map*

$$\kappa : \mathcal{X}_{\text{res}} \longrightarrow \mathcal{T}$$

from same-scope residual candidates to a finite status object $\mathcal{T}$ such that:

- (i) *κ is constant on standing-equivalent candidates;*
- (ii) *κ separates candidates only when the separation is detected by an anchor, tensor witness, skin/identification fact, or boundary-ledger role already present in S_{res} ;*
- (iii) *κ respects MR, RPC, GCI, ATS, ABT, and SGC;*
- (iv) *κ introduces no selector, metric, empirical back-import, repair, privileged chart, or enlarged witness while continuing to claim same-scope authority.*

Definition 13.5 (Terminal status quotient). *Let*

$$q_{\text{res}} : \mathcal{X}_{\text{res}} \longrightarrow Q_{\tau}$$

be the quotient map identifying two residual candidates precisely when all admissible same-scope classifiers assign them the same status-relevant role. Let

$$\tau_Q : Q_{\tau} \longrightarrow \{\text{EXCLUDE, IDENTIFY, FORCE, Skin, EXT-FIX}\}$$

be the induced status map. The pair (Q_{τ}, τ_Q) is terminal when every admissible classifier $\kappa : \mathcal{X}_{\text{res}} \rightarrow \mathcal{T}$ factors uniquely through q_{res} .

Theorem 13.6 (Residual quotient terminality / universal property). *The terminal status quotient (Q_τ, τ_Q) is terminal among admissible same-scope residual classifiers on S_{res} . Equivalently, for every admissible classifier $\kappa : \mathcal{X}_{\text{res}} \rightarrow \mathcal{T}$, there exists a unique map $\bar{\kappa} : Q_\tau \rightarrow \mathcal{T}$ such that*

$$\kappa = \bar{\kappa} \circ q_{\text{res}}.$$

The corresponding commutative diagram is

$$\begin{array}{ccc} \mathcal{X}_{\text{res}} & \xrightarrow{q_{\text{res}}} & Q_\tau \\ \kappa \downarrow & & \downarrow \bar{\kappa} \\ \mathcal{T} & = & \mathcal{T} \end{array}$$

In particular, the canonical status map itself factors as $\tau = \tau_Q \circ q_{\text{res}}$:

$$\begin{array}{ccc} \mathcal{X}_{\text{res}} & \xrightarrow{q_{\text{res}}} & Q_\tau \\ \tau \downarrow & & \downarrow \tau_Q \\ \{\text{EXCLUDE, IDENTIFY, FORCE, Skin, EXT-FIX}\} & = & \{\text{EXCLUDE, IDENTIFY, FORCE, Skin, EXT-FIX}\} \end{array}$$

Proof. If $x, y \in \mathcal{X}_{\text{res}}$ have the same image under q_{res} , then they agree on every declared anchor, tensor witness, skin/identification fact, boundary-ledger role, transport behavior, and standing quotient relevant to residual classification. An admissible classifier that separated them would therefore introduce a distinction not detected by S_{res} ; by the AMetric boundary and standing-gate closure, that distinction is a selector, privileged presentation, or second classifier, hence inadmissible. Thus $\bar{\kappa}([x]) = \kappa(x)$ is well-defined. Conversely, if two candidates have different status-relevant quotient classes, the difference is detected by one of the declared tests or witnesses, so the quotient does not collapse a standing-bearing distinction. Since q_{res} is surjective onto quotient classes, the induced $\bar{\kappa}$ is unique. Taking $\kappa = \tau$ gives the displayed factorization of the canonical status map. $\square$

Corollary 13.7 (Residual quotient uniqueness). *Any two admissible same-scope terminal residual classifiers for S_{res} are uniquely equivalent up to admissibility-preserving isomorphism. Equivalently, if (Q_τ, q_{res}) and $(Q'_\tau, q'_{\text{res}})$ both satisfy the terminal universal property for the same fixed residual scope, then there exists a unique isomorphism*

$$u : Q_\tau \longrightarrow Q'_\tau$$

such that $q'_{\text{res}} = u \circ q_{\text{res}}$, and a unique inverse $v : Q'_\tau \rightarrow Q_\tau$ such that $q_{\text{res}} = v \circ q'_{\text{res}}$. Thus no parallel inequivalent terminal residual quotient remains inside the declared same-scope comparison class.

Proof. Apply terminality of (Q_τ, q_{res}) to the classifier q'_{res} and terminality of $(Q'_\tau, q'_{\text{res}})$ to the classifier q_{res} . This gives unique maps u and v satisfying the displayed equations. The composites $v \circ u$ and $u \circ v$ are endomorphisms through which the same quotient maps factor. By uniqueness in the universal property, they are the identity maps on their respective quotients. Hence u is an admissibility-preserving isomorphism, unique by terminality. $\square$

Definition 13.8 (Universal residual quotient). *Let*

$$\pi_{\text{res}} : \mathbf{C}_{\text{res}} \rightarrow \mathbf{Q}_{\text{res}}$$

be the quotient functor identifying exactly those residual reconstruction objects and morphisms related by admissible equivalence. The category $\mathbf{Q}_{\text{res}}$ is the universal residual quotient of the declared scope; its object-level status quotient is Q_{τ} .

Theorem 13.9 (Terminal residual quotient). *In the fixed witness fiber of $\mathbf{C}_{\text{res}}$, the residual status quotient $[\tau]$ is terminal in the following reconstruction sense: every same-scope residual reconstruction preserving S_{res} factors uniquely through $[\tau]$ up to witness-preserving representation equivalence.*

Proof. By residual status exhaustion, every candidate has one of the five statuses. By witness completion, a same-scope object preserving the witness fiber cannot alter a status without altering a witness or importing a selector. The object-level universal property in theorem 13.6 therefore applies fiberwise: every admissible reconstruction induces the same status quotient, and any classifier or reconstruction functor constant on admissible equivalence classes factors uniquely through the quotient. The categorical factorization through $\mathbf{Q}_{\text{res}}$ follows from the definition of π_{res} . $\square$

Definition 13.10 (Same-scope residual tightening functor). *A same-scope residual tightening functor is an endofunctor*

$$\mathcal{F} : \mathbf{C}_{\text{res}} \rightarrow \mathbf{C}_{\text{res}}$$

that preserves K_{SM} , W_{res} , T_{SM} , $\sim_{\text{st}}$, the boundary ledger $\mathcal{E}_{\text{SM}}$, and the no-selector/no-repair discipline, while purporting to refine the terminal classification or value assignment of at least one residual candidate.

Theorem 13.11 (Residual quotient factorization). *Every admissible same-scope residual tightening functor descends to the terminal universal residual quotient. That is, for every admissible same-scope endofunctor*

$$\mathcal{F} : \mathbf{C}_{\text{res}} \rightarrow \mathbf{C}_{\text{res}},$$

there is a unique witness-preserving endofunctor

$$\overline{\mathcal{F}} : \mathbf{Q}_{\text{res}} \rightarrow \mathbf{Q}_{\text{res}}$$

such that the quotient square commutes up to representation equivalence:

$$\begin{array}{ccc} \mathbf{C}_{\text{res}} & \xrightarrow{\pi_{\text{res}}} & \mathbf{Q}_{\text{res}} \\ \mathcal{F} \downarrow & & \downarrow \overline{\mathcal{F}} \\ \mathbf{C}_{\text{res}} & \xrightarrow{\pi_{\text{res}}} & \mathbf{Q}_{\text{res}} \end{array} \quad \pi_{\text{res}} \circ \mathcal{F} = \overline{\mathcal{F}} \circ \pi_{\text{res}}.$$

At the candidate level, the same statement is represented by

$$\begin{array}{ccc} \mathcal{X}_{\text{res}} & \xrightarrow{q_{\text{res}}} & Q_{\tau} \\ F \downarrow & & \downarrow \bar{F} \\ \mathcal{X}_{\text{res}} & \xrightarrow{q_{\text{res}}} & Q_{\tau} \end{array}$$

Moreover, $\bar{\mathcal{F}}$ and $\bar{F}$ are terminal-status preserving on $[\tau]$. Hence a functor that would nontrivially bind N_g or a residual value must either fail admissibility, collapse to skin or representation-equivalent status, or leave the fixed residual category by adding new witness or boundary structure.

Proof. Because $\mathcal{F}$ is same-scope, it preserves the skeleton, witnesses, transport behavior, standing quotient, boundary ledger, and no-selector/no-repair discipline. Therefore if C and C' are identified in the admissible residual quotient, then $\mathcal{F}(C)$ and $\mathcal{F}(C')$ are identified as well. Thus $\mathcal{F}$ is congruent with respect to the equivalence relation used to form $\mathbf{Q}_{\text{res}}$ and descends to a well-defined endofunctor $\bar{\mathcal{F}}$ on quotient classes. The displayed square is the category-level form of the universal property in theorems 13.6 and 13.9; uniqueness follows because π_{res} is quotient-surjective.

By terminality of $[\tau]$, the descended functor cannot refine the five status classes while preserving the fixed witness fiber. To change an excluded, identified, forced, skin, or externally fixed class, $\mathcal{F}$ would need either to alter the data used in the quotient, distinguish representatives already identified by the quotient, or add a new boundary/witness authority. These alternatives are respectively scope change, selector or second-classifier import, and enlarged-domain structure. A proposed nontrivial binder for N_g or for Θ_{env} would do exactly that: it would change the value or status of an externally fixed class while claiming to preserve the quotient. Hence every admissible same-scope tightening functor descends to the terminal residual quotient and is status-preserving there. $\square$

Corollary 13.12 (No parallel residual classifier). *There is no second inequivalent admissible same-scope residual classifier running parallel to τ . Any candidate classifier factors through $\mathbf{Q}_{\text{res}}$ and agrees with $[\tau]$ up to representation equivalence; by theorem 13.7, any purported terminal alternative is uniquely isomorphic to the same quotient. A classifier that does not so factor changes the comparison class.*

14 Heuristic finite witness example

The toy model is pedagogical rather than physical. It is included only to display the terminal classification logic in finite form; no empirical Standard Model content is inferred from the toy set.

Let

$$R = \{u, b, c, g, p, s, h, e\},$$

where u is unbounded multiplicity, b is bounded unforced multiplicity, c is coherence-forced multiplicity, g is generation cardinality, p is standing-bearing residual parameter value, s is skin, h is hidden selector, and e is enlarged-fixed-boundary datum.

Define the prefilter

$$\chi : R \rightarrow \{\text{same-scope candidate, rival route}\}$$

by

$$\chi(u) = \chi(b) = \chi(c) = \chi(g) = \chi(p) = \chi(s) = \text{same-scope candidate}, \quad \chi(h) = \chi(e) = \text{rival route}.$$

Then the terminal map on same-scope candidates is

$$\begin{aligned} \tau(u) &= \text{EXCLUDE}, & \tau(b) &= \text{IDENTIFY}, & \tau(c) &= \text{FORCE}, \\ \tau(g) &= \text{EXT-FIX}, & \tau(p) &= \text{EXT-FIX}, & \tau(s) &= \text{Skin}. \end{aligned}$$

The hidden selector h has no sixth status: it fails admissibility. The enlarged envelope e is not a counterexample to the old scope; it defines a new scope.

Proposition 14.1 (Toy no-sixth-status). *The finite model has no same-scope terminal status beyond EXCLUDE, IDENTIFY, FORCE, Skin, and EXT-FIX.*

Proof. The same-scope candidates are exactly $\{u, b, c, g, p, s\}$, and τ maps them into the five displayed statuses. The remaining elements are not in the domain of τ : h is an inadmissible selector route and e is a scope change. Therefore a sixth status would require either admitting h as a selector or treating e as old-scope. Both violate the prefilter. $\square$

15 Route exhaustion

The route ledger is stated as a theorem-level audit object inside the declared same-scope comparison class, under admissibility-preserving operations and without witness enlargement. It is not a classification of enlarged-domain physics in general. A critical same-scope counterexample must specify its required extra structure and identify which closure theorem it escapes while preserving S_{res} . The ledger records standard route families, their required extra structure, and their terminal images under the fixed-scope residual quotient.

Rival route	Required extra structure	Failure mode	Closure invoked	Terminal classification
Unbounded copy of an admitted structure	Unlimited repetition operator with no terminating witness	Enlarges representation without declared coherence force; witness-inert copies quotient away	theorem 8.2	EXCLUDE or scope change
Bounded finite copy number	Finite selector for copy count	Discretionary bound is not forced by GCI; witness-inert copies identify	theorem 8.4	IDENTIFY or selector failure
Coherence-required multiplicity	Declared global witness requiring non-unity	Not a residual freedom; retained as tensor content	theorem 8.5	FORCE
Force exactly three generations internally	Generation selector or already-declared boundary datum	Selector absent from W_{res} ; if value is declared, it is externally fixed data	theorems 9.5, 9.6 and 9.8	selector failure / EXT-FIX

Collapse generation count to one	Reinterpret family cardinality as same-interior copy number	Changes flavor slots, mixing dimensions, and boundary ledger	theorem 9.8	scope change
Add extra generations as same-scope	New envelope load or hidden family labels	Witness-bearing additions change scope; witness-inert additions are redundant or excluded	theorems 8.2 and 9.5	scope change / EXCLUDE
Dualize or replicate internal family sectors	Mirror/dual sector rule or sector-replication operator	Witness-inert dualization is multiplicity encoding or skin; witness-bearing dualization changes scope	theorems 9.6 and 9.7	skin / IDENTIFY / scope change
Bind one Yukawa eigenvalue internally	Selective numerical binder	Uses empirical value, boundary datum, hidden relation, or privileged selector	theorems 10.5 and 10.10	EXT-FIX / failure
Bind a coupled flavor or parameter manifold	Collective relation on Θ_{env}	If witness-forced, already Θ_{force} ; otherwise hidden selector, boundary import, or scope change	theorems 10.7 and 10.9	FORCE / EXT-FIX / failure
Derive gauge coupling raw number	Privileged chart, scheme, or normalization	Confuses presentation coordinate with standing invariant unless additional tensor data are added	theorem 10.10	Skin or EXT-FIX
Use RG flow or fixed point as value selector	Boundary condition, scale anchor, fixed-point selection rule	Flow gives trajectories; unique values require boundary/tensor data or chart privilege	theorems 10.9, 13.6 and 13.11	boundary import / failure
Use naturalness, typicality, probability, measure, or anthropic ranking	Metric, weighting, threshold, or evaluation order	Adds boundary authority not present in the AMetric fixed scope	theorem 11.4	inadmissible selector
Use empirical value as derivation	Observation-to-principle promotion	Calibration may instantiate a slot; it does not become same-scope derivation	theorem 11.4	EXT-FIX
Hidden symmetry binds value	New symmetry witness or witness-inert relabeling	Witness-changing symmetry is enlarged scope; witness-inert symmetry is skin; selective use is selector	theorems 7.3 and 12.2	scope change / skin / failure
Alternative flavor basis counted as structure	Basis privilege	Basis change is skin unless invariant flavor witnesses change	theorem 10.10	Skin or scope change
Extra CP-like local phase	Local phase convention or new invariant holonomy	Convention is skin; invariant addition changes witness/envelope	theorems 7.3 and 10.10	Skin / EXT-FIX / scope change
Add sterile sector	New field-role slot and boundary data	Adds load/witness/envelope content unless entirely inert	theorem 2.4	enlarged scope or skin
Add extra gauge factor or Higgs multiplet	New gauge/load/tensor witnesses	Changes skeleton and envelope load	theorems 2.4 and 7.3	enlarged scope
BSM/UV completion as counterexample	New anchors, tensor witnesses, boundary laws, or projection map	Either projects to old quotient or changes load/witness/boundary data	theorems 2.4 and 13.11	factorization or scope change
Latent hidden label	Standing for unobserved label without witness	No standing without witness; with witness, tensor ledger changes	theorem 7.3	skin / scope change / failure

Post-hoc repair of failed derivation	Later filtering or tuned exception	Retroactive filtering cannot create standing in old lineage	theorem 11.4	repair failure
Canonical representative picker	Privileged representative relation	Treats skin as tensor or as boundary authority	theorem 6.1	selector failure
Partial operator totalization	Extension of partial map to total binder	Same-scope totalization is representation or inadmissible; nonconservative extension changes scope	theorems 12.5 and 13.11	representation-equivalence / failure
Future unnamed principle	Deferred authority not in present witness package	If same-scope, already an operator on current quotient; if new, enlarged-domain physics	theorems 12.5 and 13.11	exhausted / scope change

Theorem 15.1 (Theoremized route-exhaustion ledger). *Every audited route that purports to avoid residual classification while preserving S_{res} has a ledger image above. No audited route survives as an inequivalent admissible same-scope residual tightening without either factoring through the terminal quotient, collapsing to skin/identification, failing admissibility, or changing scope.*

Proof. A rival must alter at least one of the following: skeleton, witness family, transport relation, standing quotient, role typing, terminal status map, envelope boundary, or no-selector/no-repair discipline. If none changes, the rival is representation-equivalent or skin. If a witness or tensor relation changes, the proposal is a different scope. If a standing-bearing distinction is added without a witness, it is a selector or hidden label. If a numerical value is obtained by measurement, boundary matching, or larger-domain input, it is boundary import or enlarged-domain derivation. If a failed derivation is saved by later filtering, it is repair. The ledger enumerates these route families with their required extra structure, failure mode, theorem dependency, and terminal classification. Residual quotient factorization further shows that any route still claiming same-scope status must factor through the already constructed terminal quotient. Therefore no same-scope escape route in the declared route ledger remains; routes that derive additional physics by adding witnesses or boundary structure are instead enlarged-domain routes covered by theorems 2.3 and 2.4. $\square$

16 Expanded constraint ledger

Candidate	Role class	Tests applied	Failure / retention condition	Status
Unbounded multiplicity	Tensor-expanding without witness	MR, RPC, GCI	Enlarges space without coherence force	EXCLUDE
Bounded unforced multiplicity	Redundant or selector-dependent	MR, RPC, ATS	No witness requires finite bound	IDENTIFY / EXCLUDE
Coherence-forced multiplicity	Tensor	GCI, MR	Declared witness package requires non-unity	FORCE

Gauge/presentation copies	Skin	ATS	No standing-bearing witness changes	Skin / IDENTIFY
Generation cardinality	Fixed-boundary anchor	MR, RPC, ABT, SGC	Required to instantiate envelope; no same-scope binder	EXT-FIX
Gauge coupling chart number	Skin unless invariantized	ATS, ABT	Raw number changes under redescription	Skin or EXT-FIX
Gauge coupling value slot	Fixed-boundary parameter	MR, RPC, ABT	Standing-bearing instantiation value not same-scope forced	EXT-FIX
Yukawa eigenvalue/mass slot	Fixed-boundary parameter	MR, RPC, ABT	Numerical value not bound by same-scope witness	EXT-FIX
Mixing-angle/phase invariant residue	Fixed-boundary parameter after quotient	ATS, MR, RPC	Basis skin removed; invariant value not internally bound	EXT-FIX
Basis, gauge, scheme, normalization representative	Skin	ATS	Representative changes no tensor witness	Skin / IDENTIFY
Hidden selector	Not admitted residual	ABT, SGC, RPC	Selects without witness authority	inadmissible
Empirical calibration	Boundary import	MR, RPC	May instantiate value; not internal derivation	EXT-FIX
BSM/UV addition	Enlarged scope	scope lock	Changes load, witness, or envelope unless quotient-inert	scope change

17 Objections and replies

Objection	Closure reply
“You did not derive three generations.”	Correct. The theorem does not claim to. It proves that the generation cardinality is externally fixed at the boundary of the fixed Standard Model residual scope and cannot be selected by a same-scope binder.
“Then generation count is arbitrary.”	No. Its role and boundary location are fixed. Only its numerical value is not same-scope derivable here. Arbitrary skin would change no envelope ledger; N_g changes the instantiated family and parameter structure.

“You are hiding free parameters in the boundary ledger.”	The boundary classification is explicit. Parameters are partitioned into skin, forced relations, and irreducible fixed-boundary inputs. The paper states the boundary classification rather than concealing it.
“A UV theory might derive the values.”	It may. That would supply new anchors or tensor witnesses and belongs to an enlarged scope. If it projects back to the old Standard Model envelope, it is compatible with this theorem.
“Then the paper is anti-BSM.”	No. The enlarged-domain compatibility proposition explicitly allows new physics to derive fixed-boundary data. It only denies that such derivations are same-scope tightenings of the old residual quotient.
“Maybe parameters are jointly constrained even if individually unbound.”	The collective parameter non-closure theorem handles this: a coupled relation is admissible only if already witness-forced, pure skin, or supplied by enlarged-domain/boundary structure.
“Empirical values should count as derivations.”	Empirical calibration can instantiate or boundary-match an envelope slot. It does not become an internal same-scope derivation unless the value follows from declared same-scope witnesses.
“A hidden symmetry could bind the values.”	If the symmetry changes witnesses, the scope changes. If it changes no witness, it is skin. If it selects despite no witness change, it is an inadmissible selector.
“A naturalness or probability principle could pick the values.”	Metrics, measures, typicalities, rankings, and proximity structures are not boundary authority in the AMetric fixed scope. They must be declared as additional structure, hence scope change.
“The classification is semantic relabeling.”	The theorem blocks concrete moves: unbounded repetition, unforced bounded multiplicity, generation selectors, raw chart-value promotion, empirical back-import, hidden labels, and totalization repairs.
“The theorem assumes the Standard Model.”	It assumes the fixed Standard Model representation envelope as a declared downstream scope. It does not pretend to derive the whole envelope from no premises in this paper. Its job is residual exhaustion after that envelope is fixed.

“Why not collapse all multiplicities to unity?”	Only same-interior unforced repetition collapses. Envelope cardinality such as generation count is not a redundant copy operator; changing it changes the envelope.
“Why is externally fixed not a sixth status?”	Externally fixed is one of the five terminal statuses. The excluded sixth status is a residue that is neither skin, forced, excluded, identified, nor externally fixed while still claiming same-scope standing.
“This prevents future work.”	It prevents only silent same-scope tightening without new witnesses. Larger-domain work is explicitly allowed and classified as enlarged-scope derivation or boundary matching.
“A future principle could internalize the envelope later.”	Only by changing the case distinction. If it preserves S_{res} , it is already an operator on the present residual quotient and is exhausted. If it adds new authority, it is enlarged-domain physics.
“Is this ordinary underdetermination?”	No. Ordinary underdetermination leaves multiple admissible options open in the same comparison class. Here each audited residual route is assigned a terminal status by quotient, witness, and boundary tests; alternatives that survive do so only as externally fixed data or enlarged-domain physics.
“Does classification rigidity imply physical inevitability?”	No. Theorem 2.3 separates the fixed-scope rigidity theorem from claims about all physically conceivable ultraviolet or enlarged-domain completions.

18 No further same-scope residual tightening

The final theorem collects the residual-exhaustion results into a single fixed-scope statement.

Theorem 18.1 (No further same-scope residual tightening). *After the Standard Model representation envelope, representation-family closure, charge-lattice rigidity, and prior parameter-tightening stages are fixed, no admissible same-scope operation remains that can eliminate, identify, force, or internally bind any additional residual in the declared comparison class without one of the following occurring:*

- (i) *the operation is representation-equivalent or skin;*
- (ii) *the residual was already forced by the declared witness package;*
- (iii) *the operation imports a hidden selector, metric, typicality rule, empirical value, or repair;*

(iv) the operation changes the witness package, transport relation, standing quotient, role typing, or envelope boundary;

(v) the operation is a deferred-internalization claim and is therefore either already exhausted as a same-scope operator or enlarged-domain physics.

Consequently, unbounded multiplicity is excluded, unforced multiplicity identifies away, coherence-forced multiplicity is retained, presentation residue is skin, and generation cardinality together with irreducible standing-bearing numerical parameters are externally fixed and fixed-boundary irreducible within S_{res} .

Proof. Let B be a proposed further tightening operation. If B acts on a residual candidate inside S_{res} , then residual status exhaustion assigns the candidate one of the five terminal statuses. For excluded, identified, forced, and skin residuals, no further nontrivial tightening exists by definition of the terminal status and by witness completion. For externally-fixed residual data, fixed-boundary irreducibility, the no-hidden-binder theorem, and residual quotient terminality and residual quotient uniqueness imply that a further binder must add witnesses, select without witnesses, import external data, repair a failed route, or change scope. If B is postponed as a future principle, the no-deferred-internalization theorem classifies it as already an admissible same-scope operator or as enlarged-domain structure. If B acts outside the candidate space, it is a rival route and is exhausted by the theoremized route ledger: representation-equivalent, inadmissible, or scope-changing. Therefore B cannot produce a new same-scope residual tightening. $\square$

Corollary 18.2 (Fixed-scope parameter-tightening closure). *This result closes the Standard Model parameter-tightening arc in the fixed-scope residual-classification sense only: it does not derive every numerical value; it proves that each residual in the declared same-scope candidate space has terminal structural status and that no admissible operation preserving S_{res} remains to bind those residuals further.*

19 Conclusion

The result is a fixed-scope residual-exhaustion theorem for the declared Standard Model representation regime. Its content is not the prediction of every observed numerical value. Its content is the classification of every declared same-scope residual after the upstream representation skeleton, representation-family closure, charge-lattice rigidity, and parameter-normal-form constraints are fixed.

The proof has six layers. First, the residual partition theorem assigns each same-scope residual to one of five statuses:

EXCLUDE, IDENTIFY, FORCE, Skin, EXT-FIX.

Second, the hidden-selector and repair-prohibition results show that a value cannot be made internal by adding an unlicensed selector, empirical back-import, metric privilege, typicality rule, or post-hoc filter. Third, the generation-reconstruction and collective-parameter results show that family cardinality and residual numerical values cannot be bound by same-scope reconstruction, coupled-parameter constraints, RG/fixed-point language, or sector replication unless additional witness or boundary structure is supplied. Fourth, the residual quotient has a terminal universal property: admissible same-scope classifiers and tightening functors factor uniquely through it. Fifth, the theoremized route ledger shows that standard residual-escape routes either factor through the quotient, collapse to presentation or identification, fail admissibility, or change scope. Sixth, the enlarged-domain compatibility result states that a broader physical theory may still derive or explain these values by enlarging the comparison regime.

The terminal classification is compact:

unbounded multiplicity	$\mapsto$	EXCLUDE,
bounded unforced multiplicity	$\mapsto$	IDENTIFY or EXCLUDE,
coherence-forced multiplicity	$\mapsto$	FORCE,
presentation residue	$\mapsto$	Skin,
generation cardinality	$\mapsto$	EXT-FIX and fixed-boundary irreducible,
irreducible standing-bearing numerical values	$\mapsto$	EXT-FIX and fixed-boundary irreducible.

Generation count and residual parameter values are therefore not untyped arbitrary knobs within this theorem. They are externally fixed residual data: their roles, slots, quotient behavior, and boundary location are fixed, while their numerical values are not forced by any admissible operation preserving S_{res} . A larger-domain theory may attempt to derive them; if it succeeds, it will have supplied new boundary, tensor, transport, dynamical, ultraviolet, or flavor-theoretic structure. That possibility is compatible with the theorem. What is closed is the fixed-scope claim that a hidden admissible same-scope operation remains available inside the already fixed Standard Model representation regime.

The result should therefore be read as classification rigidity, not as metaphysical inevitability or ontological finality. It is also not ordinary underdetermination: the same-scope alternatives are not merely left open, but are partitioned by terminal quotient status. A same-scope counterexample must specify a new admissible operator, a new witness structure, or a new comparison regime. The first is exhausted by the quotient and operator results; the second enlarges witness content; the third changes scope.

A Proof dependency graph

- G1. The fixed residual scope declares skeleton, redescription, transport, witnesses, standing quotient, and envelope ledger.
- G2. Same-scope competitors must preserve all components of that scope.

- G3. Role decomposition forces every residual effect into anchor, tensor, or skin.
- G4. MR / RPC / GCI classify multiplicity and coherence routes.
- G5. Witness completion blocks arbitrary new residual witnesses.
- G6. AMetric boundary closure blocks selectors, metrics, probabilities, typicalities, rankings, and canonical representatives as boundary authority.
- G7. Standing-gate closure blocks a second same-domain classifier.
- G8. Same-scope operator exhaustion blocks hidden totalization or future-binder routes.
- G9. Residual status exhaustion gives the five terminal statuses.
- G10. Multiplicity exhaustion assigns unbounded, bounded unforced, and coherence-forced multiplicity.
- G11. Generation-count analysis assigns N_g to the externally-fixed residual-data class.
- G12. Parameter taxonomy partitions Θ_{res} into skin, forced, and externally-fixed parts.
- G13. Residual quotient terminality gives the universal property through which all admissible same-scope classifiers factor.
- G14. Residual quotient uniqueness blocks parallel inequivalent terminal classifiers.
- G15. Route exhaustion verifies that all audited escape-route families collapse, fail, or change scope.
- G16. The no-further-tightening theorem follows.

B Counterexample criterion

A same-scope counterexample to the residual-exhaustion theorem must exhibit a residual reconstruction object C' satisfying all of the following:

- C1. it preserves the same Standard Model skeleton K_{SM} ;
- C2. it preserves the same residual witness package W_{res} ;
- C3. it preserves admissible redescription and transport $R_{\text{SM}}, T_{\text{SM}}$;
- C4. it preserves the standing quotient $\sim_{\text{st}}$;
- C5. it preserves the same envelope boundary $\mathcal{E}_{\text{SM}}$;
- C6. it introduces no selector, metric, typicality, repair, empirical back-import, or hidden boundary authority;

C7. it changes the terminal status of at least one residual in a standing-bearing way;

C8. it does not collapse to skin, quotient-identification, or representation equivalence.

These conditions are jointly inconsistent. Conditions C1–C6 preserve the old residual quotient and terminal-status machinery; by residual quotient factorization and uniqueness, any admissible same-scope tightening functor or terminal classifier must factor through that quotient up to unique admissible isomorphism. C7 requires changing what the quotient already classifies, while C8 blocks quotient-inert routes. Therefore a same-scope counterexample cannot be constructed without denying the fixed comparison class.

C AASC dependency map

AASC source	Function imported into this paper
Minimal Conditions for Admissible Construction	Supplies the local admissibility/standing/reference/irreversibility kernel used by the fixed residual scope.
Standing-Admissibility Identity Closure	Blocks second same-domain classifiers for generation count or parameter values.
Anchor, Tensor, and Skin The AMetric Boundary	Supplies role decomposition and role-smuggling collapse. Blocks selector, metric, ranking, typicality, probability, threshold, and canonical-representative authority.
Closure by Exhaustion for Same-Scope Operators	Blocks hidden totalization, future binder, fixed-point detour, and operator-level escape routes.
A Fixed-Domain Exhaustion Theorem	Supplies fixed-domain/same-scope discipline and fail-closed boundary mutation.
A Classification Theorem for Admissible Standard-Model Structures	Supplies occupant/envelope distinction and irreducible input certification.
Structura Ex Necessitate Standardis Modelis	Supplies fixed-load Standard Model interior closure.
Gauge-Coupling Parameters and Forced Uniqueness	Supplies raw chart-parameter/standing-invariant distinction.
Rigidity of the Standard Model Charge Lattice	Supplies the closure template: scope lock, witness completion, quotient category, hidden-selector obstruction, route exhaustion, and objection map.

D Notation

Symbol	Meaning
S_{res}	Fixed residual scope after SM-FINAL and SM-PT-1–SM-PT-3
K_{SM}	Fixed Standard Model representation skeleton
R_{SM}	Admissible same-scope redescrptions
T_{SM}	Licensed transport family
W_{res}	Residual witness package
$\sim_{\text{st}}$	Standing equivalence relation
$\mathcal{E}_{\text{SM}}$	Standard Model envelope ledger
$\mathcal{X}_{\text{res}}$	Same-scope residual candidate space
τ	Terminal status map
N_g	Generation cardinality
Θ_{res}	Residual parameter vector
MR	Maximality Remainder test
RPC	Repair-Prohibition Check
GCI	Global Coherence Intersection
ATS	Anchor/Tensor/Skin role test
ABT	AMetric Boundary Test
SGC	Standing-Gate Closure
C_{res}	Residual reconstruction category
Q_{res}	Universal residual quotient category
Q_{τ}	Terminal object-level residual status quotient
q_{res}	Quotient map from residual candidates to Q_{τ}
$\bar{\kappa}$	Unique factor of an admissible same-scope classifier through q_{res}
u	Unique admissibility-preserving isomorphism between two terminal residual quotients when both satisfy the same universal property
EXT-FIX	Terminal status for externally fixed residual data: roles fixed in the declared regime while values are not same-scope forced

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